
Suggested Teaching Guideline for
Programming Concepts PG-DITISS August 2025

Duration: 34 classroom hrs + 36 lab hrs (Total: 70 Hrs)

Objective: Develop students with the ability to efficiently automate tasks, manage data, and develop custom solutions using Python and databases.

Prerequisites: Basics of Computer Fundamentals and Database

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Learning with Python by Allen Downey

Reference: Python for Unix and Linux System Administration

Note: *Every Session includes 2 Hours of theory and 2 Hours of Lab, unless indicated Otherwise.*

A. MySQL: (10 Theory hrs +10 Lab hrs)

Session 1:

- Concepts of databases, tables, and schemas.
- Syntax and examples for creating databases.
- Best practices for organizing and naming databases.
- Syntax and examples for dropping databases.

Lab Assignment:

- Create a MYSQL user.
- Create a database add tables and then add data into those tables.
- Delete/Modify a Table or Database.

Session 2 & 3:

- Writing SELECT statements to fetch data.
- Using WHERE clause for filtering data.
- Sorting data using ORDER BY.
- Basic JOIN operations to combine data from multiple tables.

Lab Assignment:

- Write SELECT statements to retrieve specific data from a database.
- Use WHERE clauses to filter data and extract relevant information.

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- Apply ORDER BY clauses to sort retrieved data for analytical purposes.

Session 4 & 5:

- Using aggregate functions (SUM, COUNT, AVG).
- GROUP BY and HAVING clauses for grouped data.
- Subqueries and nested queries.

Lab Assignment:

- Utilize aggregate functions (SUM, COUNT, AVG) to perform calculations on dataset columns.
- Employ GROUP BY clauses to group data based on specified criteria.
- Use HAVING clauses to filter grouped data based on aggregate conditions.

B. Python: (24 Theory hrs + 26 Lab hrs)

Session 6:

- Installing Python
- Your First Python Program
- Declaring Functions
- Overview of Objects
- Indenting Code
- Native Data types

Lab Assignment:

- Try to install Python in windows and Linux (if by default not installed)
- Program on Datatypes.
- Make small program to print Hello Word
- Make a script file in Python and run that script.
- Take an input from user while running the program and print the output.
- Use for/while loop and If-else condition in program.

Session 7 & 8:

- Introducing Dictionaries
- Defining Dictionaries
- Modifying Dictionaries
- Deleting Items from Dictionaries
- Introducing Lists
- Defining Lists
- Adding Elements to Lists
- Searching Lists
- Deleting List Elements
- Using List Operators

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Lab Assignment:

- Create a dictionary, print that dictionary.
- Use for/while loop while running the program.
- Insert/delete an item in dictionary.

Session 9:

- Introducing Tuples
- Declaring variables
- Referencing Variables
- Assigning Multiple Values at Once
- Joining Lists and Splitting Strings
- Introduction to Set

Lab Assignment:

- Create a tuple, print that tuple.
- Use for/while loop while running the program.
- Insert/delete an item in tuple.

Session 10:

- Optional and Named Arguments
- type, str, dir, and Other Built-In Functions
- Object References
- Socket with Python

Lab Assignment:

- Append/insert/extend/remove/sort/reverse a list.
- Append/insert/extend/remove/sort/reverse a string.
- Use Built-in functions.

Session 11:

- Regular Expressions Using python
- Scripting in Python
- Functions and Functional Programming

Lab Assignment:

- Use regular expression to take a regular expression and search any pattern.
- Do all of these with the help of regular expression.

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- Match a single character except newline.
- Match a word.
- Match a single whitespace character.
- Match any Non-whitespace character.

Session 12: (2 Theory Hrs + 4 Lab Hrs)

- Object Oriented Linux Environment
- Classes, Objects and OOPS concepts

Lab Assignment:

- Do practice of memory management questions.
- To find out Current directory of a file
- To find out all folders & files inside given folder.
- To find out absolute path and relative path.
- To check if a file or folder exists or not.
- To create/remove a Directory.
- To check the permissions.

Session 13 & 14:

- File Handling
- Directory Access Permissions
- Controls Socket

Lab Assignment:

- To read or write a file using python.
- Print everything written in a file.
- Count, lines of the file
- Count, words of the file
- To read a web page content

Session 15:

- Libraries (eg., hashlib, shodan, datetime)
- Servers and Clients Arch
- Web Servers and Client scripting

Lab Assignment:

- Create a socket between a server and a client then analyse and monitor all activities of client.

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Session 16:

- Exploit Development techniques
- Writing plugins in Python

Lab Assignment:

- Exploit using socket programming.

Session 17:

- Exploit analysis Automation Process
- Debugging basics